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Synspec 2.01 : A universal spectrum synthesis program

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ABSTRACT

Aims. We present a brief description of the newly upgraded code SYNSPEC, which allows the calculation of synthetic spectra of stars, planets, and disks, from model atmospheres.
Methods. The code has been migrated to modern FORTRAN, and supplemented with a suite of utilities that allow users to build SYNSPEC-compatible molecular line lists from EXOMOL data, compute opacity tables for hydrodynamical simulations or radiative transfer solvers, and parallelize the calculation of large grids of synthetic spectra. The suite includes as well tools to analyze observations from low-, intermediate- and high-resolution spectrographs, and a fast chemical-equilibrium solver to evaluate what are the most relevant species in any given astrophysical environment.
Results. The paper describes the physical background, and the main changes from previous versions, including examples for a number of typical use cases.

Key words. Stars: atmospheres – Radiative transfer – Molecular data – Equation of state – Planets and satellites: atmospheres – Techniques: spectroscopic – Line: formation

1. Introduction

Synthetic spectra based on model atmospheres are routinely employed in astrophysics to interpret observations of stars, galaxies, or exoplanets. The most commonly used model atmospheres are one-dimensional (1D) plane-parallel structures, but 1D spherical and 3D hydrodynamical models are available in some cases. The assumption of Local Thermodynamic Equilibrium (LTE) is the *de facto* standard, but non-LTE models have been developed for cases of interest, albeit only in 1D due to the computational burden. Having an open-source, flexible, and easy-to-use spectral synthesis code is critical for advancing many research fields.

The computer program SYNSPEC was originally created in the early 1980's, essentially as a utility program for the stellar atmosphere code TLUSTY (later described in Hubeny (1988)), but it was soon used as a stand-alone program to generate synthetic stellar spectra for model atmospheres computed with other codes, such as those of Kurucz (1979). The first (very brief) published description was presented by Hubeny et al. (1985), and later in an obscure paper by Hubeny et al. (1994), but there was no publication dedicated exclusively to SYNSPEC until a brief description in Hubeny & Lanz (2011). Relatively detailed descriptions of SYNSPEC were presented in the series of the TLUSTY and SYNSPEC guides, in particular Hubeny & Lanz (2017, hereafter referred to as Paper I) =- version 51, and finally Hubeny et al. (2021, Paper IV) – version 54.

There are several reasons for continuing modifications and improvements of SYNSPEC. One is a significant improvement and enlargement of the available atomic and molecular databases, most notably EXOMOL[1]. But the most important motivation is presented by recent large-scale spectroscopic surveys, such as APOGEE (Majewski et al. 2017) or DESI (DESI Collaboration et al. 2026), and by the advent of ultra-high spectral fidelity instruments such as ESPRESSO (Pepe et al. 2021), that call for a fast, accurate, reliable, and universal spectrum synthesis code capable of handling departures from Local Thermodynamic Equilibrium.

At the coding level, previous versions of the code, written in FORTRAN77, faced a number of problems on modern computers, mainly related to the use of large COMMON blocks. We therefore decided to convert SYNSPEC to modern FORTRAN. The upgrade enables a more efficient handling and bookkeeping of large arrays needed to store atomic, and in particular molecular, line data needed for generating synthetic spectra of cool objects.

In this paper, we describe recent improvements of SYNSPEC inspired by the motivations listed above. The program was described in detail in the above mentioned references, but for convenience we briefly summarize the physical background and the fundamentals needed to understand the operation of the code, and the content and the structure of the input and output files. For further details the reader is referred to Paper IV. Because of a significant change in the appearance of the code, we do not continue the earlier consecutive numbering of versions (ending with 56), and adopt a new scheme, calling the present version SYNSPEC2.01. The last publicly released version will now be called SYNSPEC1.54

Besides SYNSPEC itself, we present a number of utility programs, such as the suite of routines to generate molec-

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[1] https://www.exomol.com

ular line lists in the SYNSPEC format from EXOMOL data, and a short program CHEMEQ for a fast exploration of a solution of chemical equilibrium under various conditions, and with various molecular data sets. This program is also supplemented by a Python program to visualize the results.

Another interesting addition is the Python wrapper `Synple`, which provides an easy-to-use python interface to SYNSPEC. `Synple` parallelizes the calculation of a single spectrum in shared-memory computers, splitting the spectral range. It also sets up the calculation of multiple spectra (or opacity tables) in parallel on distributed computers, and includes a framework to create spectral grids and use them to infer atmospheric parameters and abundances from observed spectra.

In order not to overburden the presentation, we split the description of SYNSPEC into two parts. The first part, the present paper, concentrates on basic concepts, and the underlying physics and chemistry of. The second part constitutes a detailed users' guide, and contains a rather technical description of the input and output files and the relevant parameters. It is presented as a separate document distributed with the program[2]. In the following, we will refer to the second part of this paper as the "Manual".

2. Physical background

From the point of view of physics, the task of SYNSPEC is very simple: solving, frequency by frequency, the radiative transfer equation for a 1-dimensional slab (a planetary or stellar atmosphere, or a segment of an accretion disk), with the depth dependence of physical parameters (temperature, density, etc.) specified by an input model atmosphere. The end result is the emergent radiation flux (or intensity) at the surface of the atmosphere, which may then be compared to an observed spectrum, with the aim to determine as many physical properties as possible of an object of interest.

Numerical methods for solving the transfer equation accurately and efficiently are well established (see, e.g. Hubeny & Mihalas 2014, Chap. 12), therefore the only issue is to properly describe and quantify the absorption, emission, and scattering coefficients.

2.1. Opacity and emissivity

To understand what needs to be supplied to, or calculated by, SYNSPEC, we write down the general expression for the absorption coefficient at frequency ν, for temperature T, and electron density $n_{\rm e}$

$$\begin{aligned}\kappa_\nu &= \sum_i \sum_{j>i} \left[n_i - n_j G_{ij}(\nu)\right] \sigma_{ij}(\nu) \\ &+ \sum_i \left[n_i - n_k G_{ik}(\nu) e^{-h\nu/k_B T}\right] \sigma_{ik}(\nu) \\ &+ \sum_k n_{\rm e} n_k \sigma_{kk}(\nu, T) \left(1 - e^{-h\nu/k_B T}\right) + \kappa_\nu^{\rm add} + \kappa_\nu^{\rm sc} \quad (1)\end{aligned}$$

where the first term represents the contribution of the bound–bound transitions, the second term the bound–free transitions (with k being the final state of the corresponding process), and the third term the free–free transitions; h an k_B are the Planck and the Boltzmann constants, respectively. The summations extends over all levels of all species. The term $\kappa_\nu^{\rm add}$ represents any additional opacity, i.e., opacity sources that are not written in terms of explicit bound–bound or bound–free transitions. The scattering part of the absorption coefficient is given by

$$\kappa_\nu^{\rm sc} = n_{\rm e}\sigma_{\rm e} + \sum_a n_a \sigma_{{\rm Ray},a} \quad (2)$$

where the first term accounts for electron scattering, and the second term for the Rayleigh scattering.

Here n_i is the population of level i, and $\sigma_{ij}(\nu)$ the cross-section for the transition $i \rightarrow j$. The term G_{ij} describes the contribution of the stimulated emission, which we need not specify here explicitly as it does not involve any additional physical parameters (Hubeny & Mihalas (2014, § 14.1). $\sigma_{\rm e}$ is the Thomson cross section, $\sigma_{{\rm Ray},a}$ is the Rayleigh scattering cross section of species a.

The expression for the emission coefficient is analogous, and does not involve any additional parameters, so we do not need to write it here explicitly. Moreover, in LTE, it is simply given by

$$\eta_\nu = (\kappa_\nu - \kappa_\nu^{\rm sc}) B_\nu + \kappa_\nu^{\rm sc} J_\nu, \quad (3)$$

where B_ν is the Planck function, and J_ν the mean intensity of radiation.

Inspection of Eqs. (1) and (2) reveals that in order to deal with opacity and emissivity in a sufficiently robust and efficient way, one needs to address the following issues:

1. **Selection of opacity sources**, i.e. the lines and continua which are expected to contribute to the final predicted spectrum, or, in other words, to give explicit meaning of the summations in Eqs. (1) and (2). A selection of continuum sources is usually straightforward and easy. There are typically only several hundreds or at most thousands of such sources, so basically all of them may be included. However, a selection of contributing lines, and the associated bookkeeping, is a delicate problem since line lists, and in particular molecular ones, may easily contain of the order of several 10^9 lines.
   For the selected opacity sources, there are two important aspects of evaluating the opacities and emissivities, namely:
   (a) **Level populations** for the levels involved in the corresponding transitions. Actually, this is the only quantity entering the opacity and emissivity that needs to be *computed* by SYNSPEC; the other components have to be merely *supplied* to SYNSPEC.
   If an input model is an LTE model, atomic and molecular level populations have to be computed by SYNSPEC because an input model atmosphere only supplies the temperature and one or two basic state parameters, either electron density, mass density, total particle number density, or gas pressure. The knowledge of the temperatures and one of the state parameter is sufficient for determining other basic state parameters, and all level populations, via the equation of state. Since this is an important topic, it will be treated in more detail in the following section.

[2] https://github.com/ivanek-hub/synspec2.01

If an input model is an NLTE model, some level populations are supplied by the model, but the populations of many levels which were not considered explicitly in the model construction, have to be treated in LTE.

(b) **Cross-sections** for the corresponding transitions. For continua this is usually easy, since there are numerous sources of data, such as TOPBASE (Cunto & Mendoza 1992). Moreover, the bound-free cross-sections have already been used in the model construction, and, for instance in the case of TLUSTY models, they are actually specified as a part of input data for generating models, and therefore they do not typically require additional effort to prepare for a SYNSPEC run.
The case of bound-bound transitions (lines) is more complicated. The cross-section (also called the line profile coefficient) involves two components: (i) an oscillator strength, or a gf-value, and (ii) a specification of the normalized profile (related to the probability of absorption in a given transition as a function of frequency). For most lines one assumes the profile is determined by three processes: natural, Stark, and Van der Waals broadening, all described by the Voigt profile with specified broadening parameters. But for some lines, notably hydrogen and helium lines, additional physical effects may play role, so that the profiles cannot simply be expressed through the Voigt profile, and typically some pre-calculated tables are necessary.

2. **Additional opacities**, and their cross-sections. SYNSPEC offers several possibilities, such as collision-induced absorption (CIA) on H_2-H_2, H_2-He, H_2-H, and $H-H$ collisions, and bound-free and free-free transitions in H^-, H_2^+, and CH and OH continuum opacity. More sources can be added if necessary, supplying the corresponding routines to SYNSPEC.
3. **Additional sources of scattering**. SYNSPEC contains the necessary data for three processes of Rayleigh scattering involving atomic and molecular hydrogen, and atomic helium. Again, more processes may be added if needed.

2.2. Equation of state and level populations

A treatment of the level populations and the equation of state (EOS) depends on whether the input model atmosphere is an LTE or NLTE model. But even in the case of an NLTE model atmosphere, many atomic energy levels that are treated in LTE. We discuss the LTE case first.

2.2.1. LTE case

The population n_i of level i of the species A (with A being and atom, ion, or a molecule) is simply given by

$$n_i = \frac{N_A}{Q_A} g_i \exp(-E_i/k_B T), \tag{4}$$

where N_A and Q_A are the total number density and the partition function of a species A, g_i is the statistical weight, and E_i the excitation energy of level i. E_i and g_i are atomic parameters, predetermined from theory or laboratory experiment, so one needs to determine N_A and Q_A. The partition functions are taken from existing sources of data, as discussed in § 3.3. To determine N_A, the crucial part is solving for the global chemical equilibrium, or, in other words, solving the equation of state (EOS).

When only atoms and ions are present, solving the EOS is quite straightforward because it only involves Saha-Boltzmann equations with well known ionization energies and partition functions. Specifically,

$$\frac{N_A^+ n_{\rm e}}{N_A} = \frac{Q_A^+}{Q_A} \beta_{\rm e} \exp(-E_{A^+}/k_B T), \tag{5}$$

where $n_{\rm e}$ is the electron density, E_{A^+} is the ionization energy, and $\beta_{\rm e}$ the electron translational partition function, given by

$$\beta_{\rm e} = (2\pi m_{\rm e} k_B T/h^2)^{3/2}, \tag{6}$$

where $m_{\rm e}$ is the electron mass. Here, the species A may also be an ion. The important point is that all chemical elements are independent, and an evaluation of the total number density of the individual ionization stages is a linear process.

However, if molecules are present, the individual number densities of atomic, ionic, and molecular species are coupled together. Specifically, the number density of a neutral molecule composed of m atoms is given by

$$N_{1,\ldots,m} = \widetilde{K}^{-1} \prod_{i=1}^{m} N_i, \tag{7}$$

where $\widetilde{K}$ is the so-called equilibrium constant[3]. An expression for number densities of charged molecules is analogous, but it involves some additional subtleties about how exactly is the equilibrium constant defined (see below).

Equilibrium constants will be discussed in more detail in § 3.4. Here we take them as known functions of T, so in order to compute the number density of any molecule (for which the equilibrium constant is known), one needs to solve the set of equations that determine the number densities of the individual atomic species. These are given through the set of particle conservation equations. For any chemical element A, the conservation equation reads

$$\sum_{k=1}^{NS} N_k \, c_k^A = N_A, \tag{8}$$

where N_k is the number density of species k, k runs through all NS atomic or molecular species containing atoms or ions of element A as one of its components. Further, c_k^A is the number of atoms A in the species k (for instance, if species k corresponds to water, H_2O, then $c_k^H = 2$, $c_k^O = 1$, and $c_k^X = 0$ for all other elements X).

The system of equations (8) for all elements forms a non-linear system, which is solved by the Newton-Raphson method.

[3] We use the notation $\widetilde{K}$ to distinguish so defined equilibrium constant with that commonly used in studies of molecules, K, which is defined through partial pressures instead of number densities. In the present case, $K = \widetilde{K}(k_B T)^{m-1}$.

2.2.2. NLTE level populations

SYNSPEC offers two possibilities of incorporating NLTE effects in the calculation of a synthetic spectrum, namely (a) to adopt NLTE level populations from an input NLTE model atmosphere, or (b) to use an LTE model atmosphere and to supply a file that contains b-factors (NLTE departure coefficients) for a set of chosen levels.

When using a NLTE model atmosphere, one needs to make a proper association between the atomic energy levels that were considered in the model atmosphere construction (only a small subset of levels, or sets of several levels grouped together), and those referred to in a line list (actual energy levels). This was discussed in some detail in Paper IV. Here we present a brief outline, and more details are covered in the Manual.

In the standard case, SYNSPEC associates the energy levels exclusively by means of their energies. If additional information is available, then the program uses not only the energy, but also the quantum numbers S and L, as well as the level parity P – see below. This can be very useful for complex model atoms with many levels, or with significant uncertainties in their energies. In this second scenario, there are two necessary changes in the input data, (i) changes in the standard atomic line list, and (ii) changes in the atomic data files.

As to item (i), we have already updated the atomic line list to contain 6 additional parameters for each line, namely, quantum numbers $2S+1$, L, and P, for the lower and upper level of the transition. As to item (ii), the analogous changes were made in a number of atomic data files (distributed along the program itself), which contain eight additional parameters for each explicit energy level, namely the lower and upper limits of the level energy, and lower and upper limits of the values of three quantum numbers, $2S+1$, L, and parity. The notion of "limits" is invoked because an individual "level", as considered in an atomic data file may in fact be a superposition of several genuine levels. For a genuine level, the upper and lower limits coincide. We note that this scheme can be used for atoms from which the LS approximation is not valid – it will work as long as the quantum numbers are consistent between the energy levels in the line list and in the atomic data files.

The populations of levels with energies higher than the energy of the highest NLTE level whose population is supplied by the model atmosphere, $E_{\rm last}$, are considered to be in LTE with either the "last" level (for $E < (E_{\rm last} + E_{\rm ioniz})/2$, or with the ground state of the next ion (otherwise).

A subtle point arises when SYNSPEC invokes molecules for a NLTE input model atmosphere. The standard EOS procedure, as described above, would treat the total number densities of neutral atoms and ions as in LTE. This would, strictly speaking, be inconsistent. Therefore, the following procedure is adopted: Instead of expressing the ratio of number densities of ions to neutrals through the Saha equation, Eq.(5), this is replaced by the actual ratio of the number densities of ions to neutrals coming from the input model atmosphere. This procedure would not be necessary if neutral atoms dominate over ions, since ions would contribute very little to the total number of nuclei of the given species, but may be important for atoms with low first ionization potential, such as Na, Al, Ca, and others. We feel that this procedure as consistent as possible within the framework of SYNSPEC, since this program is not designed to *perform* NLTE calculations, only *use* them as they are given by the results of an independent code employed to obtain the input model atmosphere.

3. Dealing with atomic and molecular data

While SYNSPEC is quite demanding as to the number, kind, and quality of atomic and molecular data is concerned, all that is essentially transparent to the user. For all atomic and molecular processes there are always corresponding data files, distributed along with the program.

In some cases, there are several variants of input data (for instance, atomic and molecular partition functions, equilibrium constants, etc.). In these cases, the user can make choices of a particular variants of data using specific keyword parameters, described in the Manual. In any case, the default values are those which we feel provide the most accurate representation of the given dataset. Consequently, if these keyword parameters are not specified, the user can be confident that the best data available to us are being used.

The reason why we include several possibilities of treating certain atomic and molecular data in this way is that it may be of interest to compare results from various datasets.

There are several types of atomic/molecular data, which we very briefly describe below.

3.1. Basic atomic structure data

For each atom/ion treated explicitly there exists a specific file that contains the necessary atomic data. These files were constructed for TLUSTY, but they are also used by SYNSPEC. Each such file is composed of three groups of data: (i) data for on energy levels (energies, statistical weights, etc.); (ii) data for bound-free transitions (cross-sections, or pointers to internal routines that calculate them); and (iii) bound-bound transition (line) data. The latter group is needed for TLUSTY, as lines have significant influence on the model atmosphere structure, but in the context of SYNSPEC they are disregarded because all information about lines (oscillator strengths; parameters for treating the line broadening) is passed to SYNSPEC by means of the line lists (level energies, gf-values; line broadening parameters), and possibly by specific keyword parameters that point to certain non-standard treatments of line broadening.

The files are supplied along with the program, so the user does not need to construct them. Most of the data for hot stars were constructed and described by Cunto & Mendoza (1992), Lanz & Hubeny (2003, 2007), and those for cooler stars by Allende Prieto et al. (2003) and Osorio et al. (2020). However, if more extended atomic data files are required, Paper I presents a detailed guide on how to build them.

We also stress that in these files the term "energy level" does not necessarily mean a true energy level in the atomic physics sense, but it is often in fact a collection of two or more true levels, for instance fine-structure components. In some cases, in particular for iron-peak elements, the "levels" referred to in the atomic data files are in fact "superlevels" (see, e.g. Hubeny & Lanz 1995), which are a collection of a large number of levels. The superlevels are defined by the excitation energy limits, and by the parity. If superlevels are

a part of an atomic data file, as they are for some instances of model atoms for Fe I-VII, Ni I-VII, SYNSPEC needs an additional input file named `superlevels` that contain information about how is a superlevel set up. This file is also contained in the standard distribution.

Atomic data files are available for the following ions: H I, He I-II, C I-V, N I-V, O I-VI, Ne I-IV, Na I-II, Mg I-II, Al I-V, Si I-IV, P I-V, S I-VI, Ar I-IV, Ca I-II, Fe I-VII, and Ni I-VII. It should be stressed that if an atomic data file is not available, it does not mean that it cannot be handled by SYNSPEC. To clarify, the use of an atomic data file for an atom/ion by SYNSPEC has two effects: (i) its bound-free and free-free opacity (continua) are allowed to contribute to the total opacity in SYNSPEC, and (ii) if the input model atmosphere is an NLTE model, then SYNSPEC adopts the NLTE populations of levels included in the atomic data file. In the latter case, a careful determination of their association to the energy levels referred to in SYNSPEC through the line list is needed. The populations of other energy levels, referred to in the line list and not associated with the levels in the atomic data files, are treated in LTE.

3.2. Data for line broadening

There are specific files with tables of line broadening data, mostly for hydrogen, since hydrogen lines are often critically important for synthetic spectra. Special line broadening tables are also available for He I and II. The corresponding keyword parameters are presented in the Manual.

At present there are no special data for other lines. The broadening of all metal and molecular lines is described through the Voigt profile, where the broadening parameters (damping constants) are specified in the line list or by specific keyword parameters, described in the Manual, Therefore, no additional files are required.

3.3. Partition functions

In the initial versions of SYNSPEC, the partition functions of atoms and ions were computed by analytical expressions from Traving et al. (1966), with all necessary parameters being hardwired in the code. Later, the treatment was upgraded by using Irwin (1981) data, expressing the partition functions for both atoms and molecules as a polynomial expansion. More recently, the default treatment for atoms and molecules is to use Barklem & Collet (2016, hereafter BC) data. By means of a specific keyword, the user can still choose the old Irwin, or even Traving et al., data for atoms, if desired.

For molecules, there is a choice between three sources of data for partition functions:

- Tsuji (1973) extensive data set, adopted if no other source of data for a given molecule exists.
- Irwin (1981) from the original Irwin table, available for 324 molecular species.
- BC data. They are implemented in SYNSPEC in two possible ways. The first one is using the updated Irwin table that includes additional data based on BC tables, transformed to a polynomial expansion analogous to that of Irwin (containing 500 molecular species). A second possibility is to interpolate directly in the BC tables.
- For a few molecules, partition functions are available from EXOMOL. At present, these are implemented in SYNSPEC for the following 29 diatomic molecules:
  AlO, C_2, CH, CN, CO, CS, CaH, CaO, CrH, FeH, H_2, HCl, HF, MgH, MgO, N_2, NH, NO, NS, NaH, OH, PH, SH, SiH, SiO, SiS, TiH, TiO, VO;
  and 3 triatomic molecules: H_2O, H_2S, and CO_2.

There is a caveat when using the EXOMOL data. The EXOMOL tables contain values of the partition functions for temperatures between $T = 1\,\mathrm{K}$ and $T = T_{\max}$ with a step of 1 K. The value of $T_{\max}$ varies for the individual species; ranging from 10,000 K (for H_2O, H_2, C_2, and FeH) down to 3,000 K (for CN, CS, CaH, CrH, MgH, and H_2S), and being typically 9,000, 8,000 or 5,000 for the rest. In order to provide values for $T > T_{\max}$, we use the following expression

$$Q(T) = Q^{\mathrm{BC}}(T)\frac{Q^{\mathrm{EXO}}(T_{\max})}{Q^{\mathrm{BC}}(T_{\max})}, \quad \text{for} \quad T > T_{\max}, \tag{9}$$

which scales the BC values at $T > T_{\max}$ to match the EXOMOL data at $T_{\max}$.

There are special keyword parameters that can be used to choose which source of data is adopted. By default, the selection is performed from the following sources, in increasing order of priority:

1. Tsuji data;
2. upgraded Irwin/BC data, or (if chosen) BC tabular data;
3. EXOMOL data, possibly scaled using Eq. (9).

In some cases, the BC and isotope-averaged EXOMOL partition functions agree well, but in some cases they do not. In Fig. 1 we show three typical cases: excellent agreement (for CO), good agreement (for CH), and significant disagreement (for CN).

3.4. Equilibrium constants

3.4.1. Neutral molecules

We generally use the formalism of Tsuji (1973). For illustration. we take here an example of a molecule composed of three types of atoms, each with an arbitrary number of nuclei. The partial pressure of a molecular species $A_iB_jC_k$ formed by i nuclei of atom A, j nuclei of atom B, and k nuclei of atom C, is given through the partial pressures of atoms A, B, and C, and the equilibrium constant K as

$$K = \frac{P_A^i P_B^j P_C^k}{P_{A_iB_jC_k}}, \tag{10}$$

where K is the equilibrium constant, given by

$$K = \frac{\beta_A^i \beta_B^j \beta_C^k}{\beta_{A_iB_jC_k}} \frac{Q_A^i Q_B^j Q_C^k}{Q_{A_iB_jC_k}} (k_BT)^{i+j+k-1} \exp(-D_0/k_BT) \tag{11}$$

where β is the translational part of the partition function,

$$\beta_A \equiv (2\pi M_A k_B T/h^2)^{3/2}, \tag{12}$$

Q is the partition function, M the atomic mass, D_0 the dissociation energy, P the partial pressure, and T the temperature. In the original Tsuji formalism, which considers

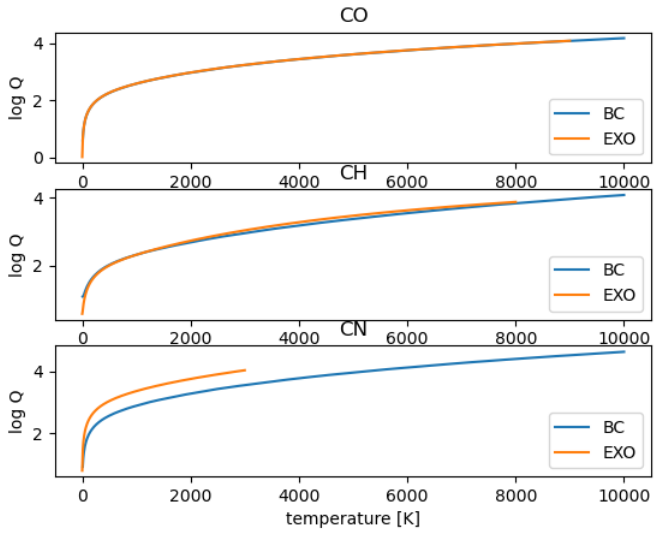


Fig. 1: A comparison of the BC and EXOMOL partition functions. The upper panel shows an excellent agreement for CO; a quite good agreement for CH (middle panel), and a disagreement for CN (lower panel). The label EXO indicates the EXOMOL partition function, averaged over isotopologues, while BC indicates the Barklem & Collet (2016 - BC) partition functions. The figure also illustrates an unfortunate feature of the EXOMOL data, having different upper temperature limits for different molecules.

only neutral and negatively charged molecules, electrons are formally considered as one of the atomic species.

Although the equilibrium constant can be determined using Eq. (11), it is easier for applications to use existing tables or polynomial fitting coefficients. One possibility is to use Tsuji's expression

$$\log_{10} K = \sum_{i=0}^{4} a_i \theta^i, \tag{13}$$

where $\theta = 5039.9/T$. Tsuji presents coefficients a_0 to a_4 for 338 molecular species, and also several negative ions such as H^-, F^-, C^-, O^-, and others, which are treated as a two-component molecule. As reported in the previous release of SYNSPEC(version 1.54; Hubeny et al. 2021), the original Tsuji table was extended by providing analogous fits for tabular values of a number of diatomic molecules and their ions computed by BC. This updated Tsuji-like table contains data for 503 molecules.

A new possibility now implemented in SYNSPEC is to use detailed tabular values of BC, namely their Table 7, and to interpolate to values of temperature of interest. Such an approach is more accurate, in particular for very low temperatures, because the BC tabular values extend down to 10 K, while the polynomial fitting formula was constructed for temperatures higher than 1000 K. Also, detailed BC tabular values are needed for molecular ions, for which the fitting formula provided inaccurate or even erroneous results. There is yet another possibility, namely using the definition expression (11) with the EXOMOL data for partition functions, and the dissociation energies from BC, but this approach has not yet been implemented in SYNSPEC or CHEMEQ.

To summarize, SYNSPEC uses the following default approach, analogous to that adopted in CHEMEQ with default settings of the keyword parameters, namely:

- for all molecules, the equilibrium constant is determined using Eq. (13), with the coefficients $a_0, \ldots, a_4$ which are given by the modified Tsuji-like table, and used in recent versions of SYNSPEC, starting with version 54, as described in Hubeny et al (2021).
- If the parameter IEQBC is set to 1 (a default in both programs), the new default treatment (although it still can be switched off if required), the equilibrium constant is evaluated by an interpolation in the BC Table 7, for all 291 molecules for which BC provides data.

Numerically, when using the BC equilibrium constants, which are expressed in SI units, they should be multiplied by 10, because TLUSTY and SYNSPEC work in the cgs units, and K for diatomic molecules has the same dimension as pressure.

3.4.2. Charged molecules

In the BC formalism, the molecular ion AB^+ is understood as a molecule with components A and B^+. Notice also that BC specify in their Table 1 which of the components is an ion. This actually makes a significant difference in the numerical value of the adopted D_0. Let us assume here that the ion is B^+. Equation (11) for this molecule thus reads

$$K^{\rm BC}_{AB^+} \equiv \frac{P_A P_{B^+}}{P_{AB^+}} = \frac{\beta_A \beta_{B^+}}{\beta_{AB^+}} \frac{Q_A Q_{B^+}}{Q_{AB^+}} (k_B T) \exp(-D_0^+/k_B T), \tag{14}$$

where we use the label BC to stress that the equilibrium constant is defined through the BC formalism.

TLUSTY and SYNSPEC adopt the Tsuji (1973)-type formalism, in which the partial pressure for a molecule or its ions is expressed through the partial pressures of the *neutral atoms*, which is advantageous from the point of solving the EOS - see also Eq. (7). Therefore, the equilibrium constant (identified by a superscript TS) is understood as

$$K^{\rm TS}_{AB^+} \equiv \frac{P_A P_B}{P_{AB^+} P_e}. \tag{15}$$

Expressing P_{B^+} through the Saha equation, Eq. (5),

$$\frac{P_{B^+} P_e}{P_B} = \frac{\beta_{B^+} \beta_e}{\beta_B} \frac{Q_{B^+} Q_e}{Q_B} (k_B T) \exp(-E_B/k_B T), \tag{16}$$

where E_B is the ionization energy of atom B, and thus its relation to the equilibrium constant defined by BC is

$$K^{\rm TS}_{AB^+} = K^{\rm BC}_{AB^+} \left(\frac{P_{B^+}}{P_B} P_e \right)^{-1} = K^{\rm BC}_{AB^+} \times {\rm corr}, \tag{17}$$

where

$${\rm corr} = \frac{1}{2\beta_e k_B T} \frac{Q_B}{Q_B^+} \exp(E_B/k_B T). \tag{18}$$

The last equality follows from Eq.(5), with $\beta_{B^+} = \beta_B$, and $Q_e = 2$. Equation (18) can also be expressed as

$$\begin{aligned} \log({\rm corr}) &= 0.17599 + \log(Q_B) - \log(Q_{B^+}) - 2.5 \log(T) \\ &+ 5039.9 E_B/T, \end{aligned} \tag{19}$$

where the ionization potential E_B is expressed in eV, and log means $\log_{10}$.

The situation is similar for negative ions. Equation (14) remains essentially unchanged, simply replacing superscript "+" by "−". Equation (16) is modified accordingly,

$$\frac{P_{B^-}}{P_B P_e} = \frac{\beta_{B^-}}{\beta_B \beta_e} \frac{Q_{B^-}}{Q_B Q_e} \frac{1}{k_B T} \exp(E_{B^-}/k_B T). \tag{20}$$

Therefore, in order to use a Tsuji-like expression,

$$\frac{P_A P_B P_e}{P_{AB^-}} = K^{\rm TS}_{AB^-} \tag{21}$$

which effectively treats a negative ion of a diatomic molecule as a three-component molecule, the relation between $K^{\rm TS}_{AB^-}$ and $K^{\rm BC}_{AB^-}$ has to be given by an equation analogous to (17),

$$K^{\rm TS}_{AB^-} = K^{\rm BC}_{AB^-} \left(\frac{P_B}{P_{B^-}} P_e\right) = K^{\rm BC}_{AB^-} \times {\rm corr}, \tag{22}$$

with

$${\rm corr} = 2\beta_e (k_B T) \frac{Q_B}{Q_{B^-}} \exp(-E_{B^-}/k_B T), \tag{23}$$

or

$$\begin{aligned} \log({\rm corr}) &= -0.17599 + \log(Q_B) - \log(Q_{B^-}) + 2.5\log(T) \\ &- 5039.9 E^B_{\rm aff}/T, \end{aligned} \tag{24}$$

where $E_{B^-} = E^B_{\rm aff}$, with $E_{\rm aff}$ being the electron affinity energy of atom B. The partition functions of (some) negative ions are given in BC Table 6.

4. Line lists

Line lists are at the very heart of the work accomplished by SYNSPEC, and several variants are available in the standard distribution. A complication here is that the lists may contain data for many millions of lines (atomic lists), and literally billions of lines for molecules. Therefore, some choices have to be made by the user, according to the required accuracy of the resulting spectra, and also with the consideration of the memory and time constraints on available computing resources.

4.1. Atomic line lists

The line lists were described in Paper IV, but for completeness we present a brief synopsis here. We stress that an atomic line list does not contain any entries for hydrogen lines (and if it does, they are disregarded), because all the necessary information is hardwired in the program.

Currently, there are two possible formats for atomic line lists, the *standard* one, used in the previous versions of SYNSPEC (before 1.54), and an *extended* one, which is preferable for NLTE model atmospheres because it provides a better association between atomic energy levels used in TLUSTY and SYNSPEC – see § 2.2.2.

(*i*) The **standard line list** contains one input record for each spectral line. It contains the following parameters:
– wavelength (nm)
– the numerical code of the atom and the ionization degree, using the Kurucz convention (e.g. 2.00 = He I, 6.03 = C IV, 26.01 = Fe II, etc.)
– $\log gf$, with g being the statistical weight and f the oscillator strength
– excitation potential of the lower level ($\rm cm^{-1}$)
– J quantum number of the lower level
– excitation potential of the upper level ($\rm cm^{-1}$)
– J quantum number of the upper level
– damping parameter for the natural broadening (`agam`)
– damping parameter for the Stark broadening (`gs`)
– damping parameter for the Van der Waals broadening (`gw`).

Line profiles are given by the Voigt profile, $H(a,x)$, where $x = (\nu - \nu_0)/\Delta\nu_D$, and $a = \Gamma/(4\pi\Delta\nu_D)$, with $\Delta\nu_D$ being the Doppler width, ν_0 the line center frequency, a the Voigt damping parameter, and Γ the physical damping parameter. The latter is given by

$$\Gamma = \Gamma_{\rm rad} + \Gamma_{\rm Stark} + \Gamma_{\rm vdW}, \tag{25}$$

where $\Gamma_{\rm rad}, \Gamma_{Stark}$ and $\Gamma_{\rm vdW}$ are the natural, Stark, and Van der Waals damping parameters, respectively. They are given by the input parameters `agam`, `gs`, and `gw`, respectively, that are specified in the line list. Specifically, after Kurucz (1970),

Natural broadening:

$$\Gamma_{\rm rad} = \begin{cases} 10^{\tt agam}, & \text{for } {\tt agam} \neq 0, \\ 2.67\times10^{-22}\nu_0^2, & \text{for } {\tt agam} = 0, \end{cases} \tag{26}$$

Stark broadening

$$\Gamma_{\rm Stark} = \begin{cases} 10^{\tt gs}\, n_{\rm e}, & \text{for } {\tt gs} \neq 0, \\ 10^{-8} n_{\rm eff}^{5/2}\, n_{\rm e}, & \text{for } {\tt gs} = 0, \end{cases} \tag{27}$$

where $n_{\rm eff} \equiv Z_I^2 [E_H/(E_I - E_j)]^{1/2}$ is the effective quantum number of the upper level, j, of the transition in ion I, with the excitation energy E_j and the ionization energy E_I, Z_I is the effective charge ($Z_I = 1$ for neutrals) of the ion I, E_H is the ionization energy of hydrogen, and $n_{\rm e}$ is the electron density.

Van der Waals broadening

$$\Gamma_{\rm vdW} = \begin{cases} 10^{\tt gw}\, c_w, & \text{for } {\tt gw} \neq 0, \\ 2.5\,(n_{\rm eff}^4/Z^2)^{0.4}\, c_w, & \text{for } {\tt gw} = 0, \end{cases} \tag{28}$$

where $c_w = (N_{\rm H} + 0.42 N_{\rm He} + 0.85 N_{\rm H_2})(T/10^4)^{0.45}$,and $N_{\rm H}$, $N_{\rm He}$, and $N_{\rm H_2}$ are the number densities of neutral hydrogen, helium, and the hydrogen molecule, respectively.

For more details refer to Paper IV, § 2.5.

(*ii*) The **extended line list**. The entry for each spectral line contains, in addition to the standard 11 parameters the set of 6 new parameters, namely:
– the quantum number $2S+1$ of the lower level
– the quantum number L of the lower level
– the parity P of the lower level
– analogous values for the upper level.
These parameters are explained in more detail in Paper IV § 3.6. The atomic line list provided in the current distribution, file `gfATO.dat`, or its binary version `gfATO.bin` are *extended* lists.

4.2. Molecular line lists

There are three variants of molecular line lists, namely the *standard* (classical-broadening) list; its variant called a *minimalistic* list; and the EXOMOL-*type* list, which contains additional Van der Waals broadening parameters, described in detail in Paper IV, Appendix F, and in the Manual. We describe the format for each type of lists below.

4.2.1. Standard list

Its structure is analogous, albeit simplified, to a standard atomic list. Each input record contains the following 7 parameters for each line:
– wavelength (nm)
– numerical code (e.g. 106.00 = CH, 108.01 = OH^+, 10108.00 = H_2O, etc.)
– $\log gf$, with being the statistical weight and f the oscillator strength
– excitation potential of the lower level (cm^{-1})
– 3 damping parameters as for the atomic line list

Since molecular lines are assumed to be formed in LTE, the parameters for the upper level are not needed. The largest source of uncertainty in treating lines whose data are provided by such a list is in the evaluation of the Van der Waals broadening, since Stark broadening is typically negligible under the conditions where molecules are formed in significant numbers, and natural broadening is usually well approximated by the classical formula. The Van der Walls broadening parameter is taken as for atomic lines, namely

$$\Gamma_{\rm vdW} = \mathtt{gw}\,(N_{\rm H} + 0.42N_{\rm He} + 0.85N_{\rm H_2})\,(T/10^4)^{0.45}. \qquad (29)$$

4.2.2. Minimalistic list

Its introduction was motivated by the observation that in a vast majority of molecular line lists the parameters `gs` and `gw` have the same value for all lines, and moreover for most molecular species. Therefore, for such molecules it is a waste of computer memory to store them in the line list. Instead they are assigned when initializing line data from the line list. Similarly, the natural broadening in such lists has its classical value, which again does not have to be stored in the line list. Consequently, each entry in the minimalistic line lists contain only the first four values of the corresponding entry of the standard list.

Parameters `gs` and `gw` can be specified through the specific keyword parameters (see the Manual) which have default values 3.1×10^{-5} and 1×10^{-7}, respectively. Therefore, if they are not specified in the input, the parameters `gs` and `gw` will have the same values as in usual standard lists.

4.2.3. EXOMOL-type list

This type of list contains for each line 4 additional parameters, which describe the Van der Waals broadening following Barton et al. (2017a,b) as

$$\Gamma_{\rm vdW} = 10^{-6} c k_B\,[g_{\rm H2}(T/296)^{n_{\rm H2}} N_{\rm H_2} + g_{\rm He}(T/296)^{n_{\rm He}} N_{\rm He}], \qquad (30)$$

accounting for broadening by H_2 (1st term) and He (2nd term). Broadening by H atoms is ignored. The parameters $g_{\rm H2}$, $n_{\rm H2}$, and analogous ones for He, are in this case included in the line list. More details are given in Paper IV, Appendix F.

SYNSPEC automatically recognizes which type of the list is used, and therefore no additional input parameters need to be specified.

In the previous release of SYNSPEC, described in Paper IV, there were 3 molecular line lists available: `gfMOL.dat`, `gfTiO.dat`, and `gfH2O.dat`, together with their binary versions `*.bin`. The first one, `gfMOL`, contains data for the following 18 molecules: H_2, CH, NH, OH, NaH, MgH, SiH, CaH, CrH, FeH, C_2, CN, CO, MgO, AlO, SiO, CaO, and VO, taken mostly from the Kurucz website[4]. The other two line lists, constructed from the EXOMOL data, are for TiO and H_2O, respectively, as their names imply.

A recent development is that it is possible to use many more molecular line lists for individual molecules, or a small groups of molecules. Our plan is to set up line lists for all molecules, and molecular ions, for which the EXOMOL data are available. Since it may be cumbersome, and memory intensive to made them all available for downloading, we instead present a suite of simple programs designed to extract data directly from the EXOMOL database, and to set up individual line lists in the SYNSPEC compatible format. This is described in § 4.3, using the utility programs listed in § 7.1, and described in detail in the Manual. Corresponding single-molecule line lists are available for the following molecules: AlH, AlO, CH, CH^+, C_2, CN, CO, CaH, CaO, FeH, HF, MgH, MgO, NO, NS, OH, OH^+, SiH, SiO, TiO, and VO.

One can use these single-molecule line lists to assess the influence of the individual molecules upon the predicted spectrum. As an illustration we present in Fig. 2 a set of plots that for several molecules display a ratio of the spectrum produced with all atomic lines plus lines of the given molecular species to the spectrum produced just with atomic lines.

It should be noted that EXOMOL considers the individual isotopologues of a molecular species separately, while SYNSPEC considers a given molecule as a combination of all its isotopologues with solar isotopic ratios (or any other adopted proportions). Similarly, the partition function of a molecule is given as an appropriate average over the partition function of all its isotopologues - see § 4.3. This is accomplished by the short programs PFSUM and LIST, outlined in § 7, and described in detail in the Manual. We underline the fact that the line-list generating program LIST produces a *minimalistic* list, since the detailed broadening parameters are not available in the EXOMOL data base for most molecular species, or for stellar conditions. For the species for which EXOMOL provides data for line broadening, a more sophisticated approach has to be adopted (as was done earlier for H_2O), but this will depend on the actual molecule. Such an endeavor will be considered in a future paper.

4.3. Isotope-averaging of the EXOMOL data

SYNSPEC traditionally works in terms of isotope-averaged atomic and ionic species, and isotopologue-averaged molecular species. An extension to treat all isotopes separately would be straightforward, but so far was deemed to be

[4] http://kurucz.harvard.edu/molecules

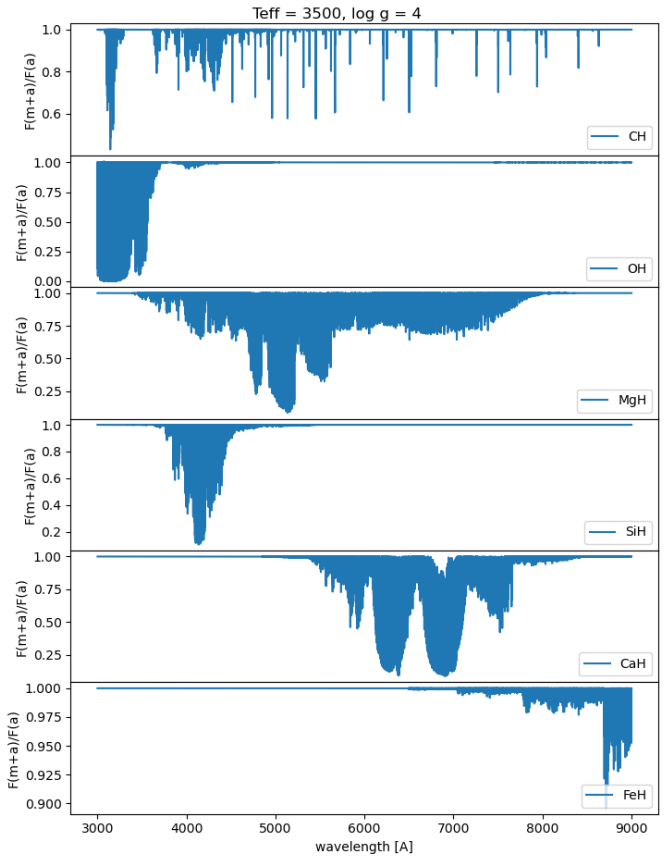


Fig. 2: The individual panels display a ratio of the spectrum produced with all atomic lines plus lines of the given molecular species, to the spectrum produced with atomic lines only, as a function of wavelengths, for a Kurucz model atmosphere with $T_{\rm eff} = 3500$K, $\log g = 4$. The panels are labelled by the name of the molecule.

unnecessary for most SYNSPEC applications. On the other hand, EXOMOL presents data with strict isotope separation, so in order to use the data in SYNSPEC, they have to be carefully averaged over the individual isotopologues, assuming, for instance, the solar isotope composition from Lodders (2020), specified in Table 1.

We consider here for simplicity a diatomic molecule, but the situation is analogous for any number of atomic constituents. The line opacity for the isotopologue Y$\equiv {}^k$A^{l}B is given by

$$\kappa_{ij}(\nu) = (gf)^Y_{ij} (N_Y/Q_Y)\, \phi^Y_{ij}(\nu, T). \tag{31}$$

Here

$$\phi_{ij}(\nu, T) = \frac{\pi e^2}{m_e c} \exp(-E_i/k_B T) \frac{H(a_{ij}, x)}{\sqrt{\pi}\Delta\nu_D}, \tag{32}$$

In SYNSPEC, the line opacity for Y is expressed as a function of the total number density for a molecule AB,

$$\kappa^Y_{ij}(\nu) = (gf)^Y_{ij} \frac{N_{AB}}{Q^{\rm ref}_{AB}} r^k_A r^l_B\, \phi^Y_{ij}(\nu, T), \tag{33}$$

where the "reference" partition function is given by

$$Q^{\rm ref}_{AB} = \sum_k \sum_l Q^{k,l}_{AB}\, r^k_A\, r^l_B. \tag{34}$$

and r^k_A stands for the fractional abundance of the isotope ^{k}A with respect to the total abundance of element A, etc. This expression is valid assuming that the atomic partition functions are isotope-independent, the dissociation constants for all isotopologues are equal, and neglecting differences in the reduced mass for the individual isotopologues.

The line opacity can then be written as

$$\kappa^Y_{ij}(\nu) = (gf)^{\rm mod}_{ij} \frac{N_{AB}}{Q^{\rm ref}_{AB}}\, \phi^Y_{ij}(\nu, T), \tag{35}$$

where the "modified" gf value is given by

$$(gf)^{\rm mod}_{ij} = (gf)^Y_{ij} \left(\frac{Q^Y_{AB}}{Q^{\rm ref}_{AB}}\right) r^k_A r^l_B, \tag{36}$$

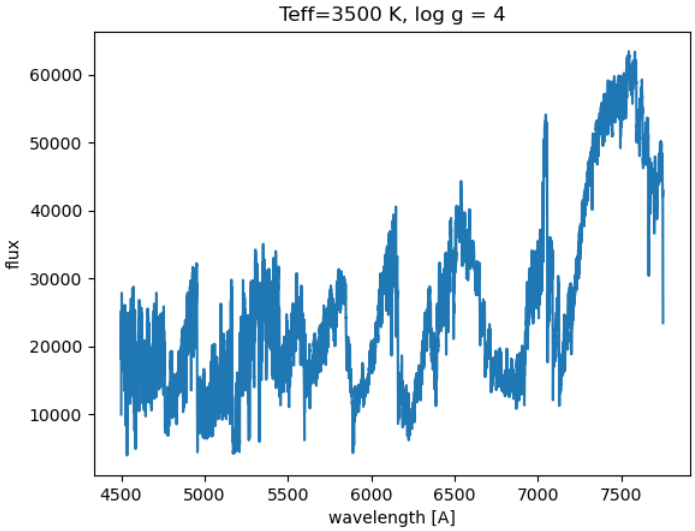


Fig. 3: Synthetic spectrum for a Kurucz model with $T_{\rm eff} = 3500$K, $\log g = 4$, computed with new molecular line lists based on EXOMOL data including AlH, AlO, C_2, CH, CH^+, CN, CO, CaH, CaO, FeH, HF, MgH, MgO, NO, NS, OH, OH^+, SiH, SiO, and TiO. For visibility, the original spectrum has been smoothed, and the plot shows the 100-points running mean.

As an example of a synthetic spectrum computed with new molecular line lists based on EXOMOL data we present in Fig. 3 a calculation for a Kurucz model with $T_{\rm eff} = 3500$K, $\log g = 4$ including the following molecules (see also the example given in the Manual): AlH, AlO, C_2, CH, CH^+, CN, CO, CaH, CaO, FeH, HF, MgH, MgO, NO, NS, OH, OH^+, SiH, SiO, and TiO, using a separate line list for each molecule.

5. Basic modes of operation of SYNSPEC

There are several modes of operation of SYNSPEC, out of which the following three are the most basic ones: computing a synthetic spectrum, constructing an opacity table, and computing depth-dependent radiative accelerations for individual atoms and ions.

Table 1: Solar isotope fractions

Label	At.#	Nuc.#	Isot.frac.	Label	At.#	Nuc.#	Isot.frac.
^{1}H	1	1	99.998	^{39}K	19	39	93.132
^{2}H	1	2	0.002	^{40}K	19	40	0.147
^{12}C	6	12	98.965	^{41}K	19	41	6.721
^{13}C	6	13	1.035	^{40}Ca	20	40	96.941
^{14}N	7	14	99.774	^{42}Ca	20	42	0.647
^{15}N	7	15	0.226	^{43}Ca	20	43	0.136
^{16}O	8	16	99.777	^{44}Ca	20	44	2.086
^{17}O	8	17	0.035	^{49}Ca	20	49	0.187
^{18}O	8	18	0.168	^{46}Ti	22	46	8.249
^{19}F	9	19	100.000	^{47}Ti	22	47	7.437
^{23}Na	11	23	100.000	^{48}Ti	22	48	73.72
^{24}Mg	12	24	78.992	^{49}Ti	22	49	5.409
^{25}Mg	12	25	10.003	^{50}Ti	22	50	5.185
^{26}Mg	12	26	11.005	^{50}V	23	50	0.250
^{27}Al	13	27	100.000	^{51}V	23	51	99.750
^{28}Si	14	28	92.230	^{50}Cr	24	50	4.345
^{29}Si	14	29	4.683	^{52}Cr	24	52	83.790
^{30}Si	14	30	3.087	^{53}Cr	24	53	9.500
^{31}P	15	31	100.000	^{54}Cr	24	54	2.365
^{32}S	16	32	95.047	^{55}Mn	25	55	100.000
^{33}S	16	33	0.744	^{54}Fe	26	54	5.845
^{34}S	16	34	4.196	^{56}Fe	26	56	91.754
^{36}S	16	36	0.013	^{57}Fe	26	57	2.119
^{35}Cl	17	35	75.765	^{58}Fe	26	58	0.282
^{37}Cl	17	37	24.235				

5.1. Synthetic spectrum

Computing a stellar synthetic spectrum is the original and standard application of SYNSPEC. Later it was extended to produce a synthetic spectrum of an accretion disk, and with the most recent updates it becomes feasible to use the code for planetary atmospheres. From the point of view of spectrum synthesis, a single ring of an accretion disk behaves exactly like a stellar atmosphere, and therefore computing its spectrum is completely analogous to the stellar case (the only small difference is in the inner boundary condition for the radiative transfer equation). The total spectrum of the disk is obtained by integrating synthetic spectra for all individual rings (see, e.g. Linnell & Hubeny 1996; Hubeny & Long 2021).

To begin with, one has to supply an input model atmosphere (or a disk ring). It is typically given as a previously computed TLUSTY model, but it can also adopt the format of a Kurucz model. One can use a model from other sources, such as the MARCS models (Gustafsson et al. (2008)) but in this case the user has to transform the model file to TLUSTY format. This can be automatically handled by the python wrapper `Synple`(see §7.3). Along with the model atmosphere one has to supply a standard input file, which contains the main control data, specifying the chemical composition, selection of atomic energy levels treated explicitly (and consequently the bound-free transitions from them being taken into account), and a specification of the free-free transitions. This file may also contain atomic data for lines that were taken into account for building a model atmosphere, but these are disregarded in SYNSPEC since it takes the information about lines from a different source (the line lists). If a model was not constructed by TLUSTY, this standard input file is set up to simulate the TLUSTY input, i.e, with the appropriate chemical composition, and bound-free and free free cross-sections.

There is a number of additional input files, discussed in detail in the Manual. The most important files are obviously the line list(s), discussed in § 4, and the input file(s) with the values of keyword parameters that set details of an evaluation of the spectrum, such as the wavelength range, spectral resolution, parameters for rejecting weak lines, and switches to turn on or off various additional processes, and switches for choosing specific variants of input atomic or molecular data. All keyword parameters are assigned default values, which represent the appropriate values for most cases, so in the actual applications only very few of them need to be explicitly set.

All the input files are described in detail in the Manual.

5.2. Opacity tables

SYNSPEC can be used for other purposes than just computing a synthetic spectrum. The issues discussed in § 2 are, from the point of view of atomic physics, *local* issues, in the sense that an opacity and the associated quantities are evaluated at a given point in the atmosphere, independently of other points. This makes SYNSPEC quite suitable for instance for generating *opacity tables*, that is, a table giving the opacity as a function of frequency (wavelength) for a number of temperatures and densities. Such a table is to be used in a subsequent model atmosphere construction This concept was discussed at length in Paper IV.

In this case, the opacity is computed exactly the same way as in the standard mode. However, there is no input model atmosphere; instead the specific input file specifies a set of temperature/density, (T, ρ), pairs (optionally, one can also use temperature/electron density, or temperature/gas pressure pairs). The values are logarithmically spaced. In the case of cool stars, having the density as the second parameter is advantageous because the electron density may become very small, and be dominated by trace elements. For hot stars, any of the state parameters can be used, but since a table may encompass both low and high temperatures, choosing density is always a safe option.

One such pair is formally considered as an "atmosphere" with just one depth point. The opacity is computed in the same way as for a normal atmosphere. SYNSPEC then loops over all (T, ρ) pairs, and stores the opacities in a specific output file. There is a subtle point here: For each (T, ρ) pair, the program has first to solve for the global chemical equilibrium to determine all number densities of all atomic and molecular species. To this end, all chemical elements must be present because all of them may in principle participate in the global chemical equilibrium.

However, the opacity table may be constructed either to include the opacity of all species, or only opacities of certain elements. The latter option is invoked in view of potential applications in TLUSTY, because it may be a very good strategy in the model atmosphere construction to treat some, important, elements in NLTE, while other elements, less important, but still contributing to the total opacity, by means of an opacity table. For instance, for computing model atmospheres of solar-type stars, one may want to include, say, H, Mg, Ca (and perhaps some other elements) in NLTE, but still consider opacities of other species, including molecular ones, in the model. This can be achieved by constructing an opacity table that excludes H, Mg, and Ca from the opacity table, but not from the global chemical equilibrium.

A possible caveat of this approach is that the EOS, and thus the formation of molecules containing the "NLTE" elements, is evaluated assuming LTE in such species. If NLTE effects cause a significant over-ionization in these species, and if ions sequester a non-negligible portion of the nuclei of the given species, the molecular abundances may be inaccurate.

SYNSPEC offers a flexibility is setting the set of (T/ρ) pairs. It can be either rectangular, or slanted rectangular in the $(\log T/\log \rho)$ space, or it can even occupy a completely irregular area in the $(\log T/\log \rho)$ space. Another option is to consider a "model tailored" opacity table. In this case, the (T, ρ) pairs are taken from an existing model atmosphere with ND depth points, and constructs an opacity table just for ND temperature/density pairs, corresponding to exact values of temperature and density of the input model, (T_d, ρ_d), $d = 1, \ldots,$ ND. Besides the temperature and density, SYNSPEC also takes the electron density from the input model, instead of evaluating it using the LTE equation of state. This table may be useful for assessing the accuracy of the numerical procedure of using pre-tabulated opacities, and for increasing the accuracy of computed models.

The setting of the opacity table is specified in a specific input file, described in detail in the Manual. That file also specifies the limiting wavelengths, the number of wavelengths in the table, and the way how the opacities are transformed from the internal opacities computed by SYNSPEC on a internal high-resolution mesh, to the wavelengths considered in the resulting table.

5.3. Radiative accelerations for radiative levitation calculations

We follow the approach of Chayer et al. (1995) who presented a useful formalism for computing non-homogeneous abundances of elements in white dwarf atmospheres in the presence of radiative levitation. This is accomplished by solving the equilibrium diffusion equation

$$g^A_{\rm rad} = g^A_{\rm eff}, \tag{37}$$

where $g^A_{\rm eff}$ is the effective local gravity acceleration, and $g^A_{\rm rad}$ is the radiative acceleration on an element A.

From the point of view of radiation quantities, the basic ingredient is the elementary radiative acceleration of an element A in its ionization stage i, given by

$$g^{A,i}_{\rm rad} = \frac{4\pi}{c}\frac{1}{X^A_i}\int_0^\infty H_\nu \frac{\kappa^{A,i}_\nu}{\rho} d\nu = \frac{4\pi}{cm_H}\frac{1}{N_{A,i}\mu_A}\int_0^\infty H_\nu \kappa^{A,i}_\nu d\nu \tag{38}$$

where H_ν is the monochromatic Eddington flux at frequency ν, $\kappa^{A,i}_\nu$ is the monochromatic opacity (per unit length) of the species A in the ionization stage i, and $X^A_i = m_A N_{A,i}/\rho$ its mass fraction. Further, ρ is the mass density, $N_{A,i}$ the number density of atoms A in the ionization stage i, m_A the mass of atom A, and $\mu_A = m_A/M_H$ the atomic weight, and m_H the mass of hydrogen atom.

Radiative acceleration of the element A, $g^A_{\rm rad}$, is given as an appropriate average of the radiative accelerations of the individual ionization stages; the corresponding expressions are given by Chayer et al. The key is that the coefficients entering the evaluation of the average radiative acceleration are given solely by the local conditions (temperature, particle number densities), and so does the evaluation of the effective gravity acceleration. SYNSPEC is therefore in an excellent position to provide accurate radiative accelerations for individual ions, which are crucial ingredients for subsequent calculations of diffusion equilibrium and thus a non-homogeneous abundance distribution in the presence of radiative levitation. Since the radiative accelerations depend on chemical abundances, one needs to iterate between the recalculation of the equilibrium abundances using Eq. (37), and the subsequent application of SYNSPEC that provides updated radiative accelerations. The corresponding program, which goes well beyond the scope of SYNSPEC, will be described in a separate paper.

5.4. Other applications

There are several more or less outdated applications, but two of them are still useful, (i) the line identification option, and (ii) the EOS-only option. Both represent a subset of calculations being done in the standard application of SYNSPEC.

– In the **line-identification** option, the program performs a selection of lines that are expected to contribute to the opacity by computing the line-center and the continuum opacities in one depth point, called *standard depth* – see the Manual. The program does not solve the transfer equation to get specific intensities, but instead computes approximate equivalent widths of lines. Its output is

an identification table, that shows approximate strength of lines. It is therefore very useful for identifying the lines in the synthetic spectrum.

The same file is produced[5] in the standard run of SYNSPEC but if the file is not produced or deleted by mistake, or if the user wants to get approximate line strengths for slightly different conditions, this option is useful because it avoids a more time-consuming recalculation of the synthetic spectrum.

– In the **EOS-only** option, SYNSPEC merely generates an output with several tables showing the results of the chemical equilibrium (EOS) calculations, without computing opacities and/or the spectrum.

However, this option does very much the same as the auxiliary program CHEMEQ, described in § 7.2. The difference between these two possibilities is that the EOS-only option produces the EOS tables for several chosen depths in the actual atmosphere, while the program CHEMEQ produces such tables for a single chosen (T, ρ) pair, but with a more comprehensive output.

Finally, we mention that there is also an option of computing spectrum for a spherical atmosphere with given velocity fields (winds), but this option was used very rarely, and although the corresponding routines are present in the source code, they were not maintained. Nevertheless, this option will be made fully operational in a future release of SYNSPEC.

6. Recent modifications of SYNSPEC

6.1. Treatment of molecules

The updated SYNSPEC version implements several improvements in the treatment of molecules and molecular line lists:

- The evaluation of the molecular partition functions and equilibrium constants has been improved with respect to the previous version. There is a new option to use directly the tabular data from BC, as well as from EXOMOL for partition functions for the molecules considered in these databases. The advantage is that one can go to very low temperatures (10 K for BC data, and 1 K for EXOMOL data), while in earlier SYNSPEC versions polynomial fits were adopted and these may give inaccurate or even wrong values of the equilibrium constants in some cases. Nevertheless, interpolating in the BC tables is far safer for any temperature, and does not lead to any appreciable increase of computer time.
- Molecular ions (charged molecules) are now treated properly, both in their correct identification in a line list, as well as in the evaluation of their equilibrium constants. This is described in detail in § 3.4.2.
- One can now consider an arbitrary number of individual molecular line lists. The advantage over the previous approach, where only up to 3 line lists were allowed, is that one can generate the individual line lists, as described in the Manual, and use them without the need to merge them into one big file, avoiding handling issues. Adopting this methodology, one can also easily explore the relative importance of individual molecular species in the resulting synthetic spectrum, as illustrated in Fig. 2.

6.2. Fixing previous bugs

Two errors were present in the previous SYNSPEC version, and have been identified and corrected in the new version. We describe them below.

A very unfortunate bug that was present in version 54 (but not in earlier ones), was an incorrect treatment of hydrogen Rayleigh scattering for hot stars. The issue was that the scattering coefficient was calculated using the total hydrogen number density, instead of the neutral hydrogen ground-state population. This does not lead to appreciable errors for cool stars since most of hydrogen is concentrated in the ground state of neutral atoms anyway, but it may grossly overestimate the influence of the Rayleigh scattering for hot stars. This is illustrated on Fig. 4 where we display a synthetic spectrum for a BSTAR07 (Lanz & Hubeny 2007) NLTE metal line-blanketed model atmosphere with $T_{\rm eff} = 20,000$ K; $\log g = 4$, computed with the current version of SYNSPEC, and with SYNSPEC, version 1.54. This is partially mitigated by the (usually more important) contribution of other opacity sources, such as the Balmer continuum of hydrogen and/or the Lyman lines, but users need to be aware of this previous problem.

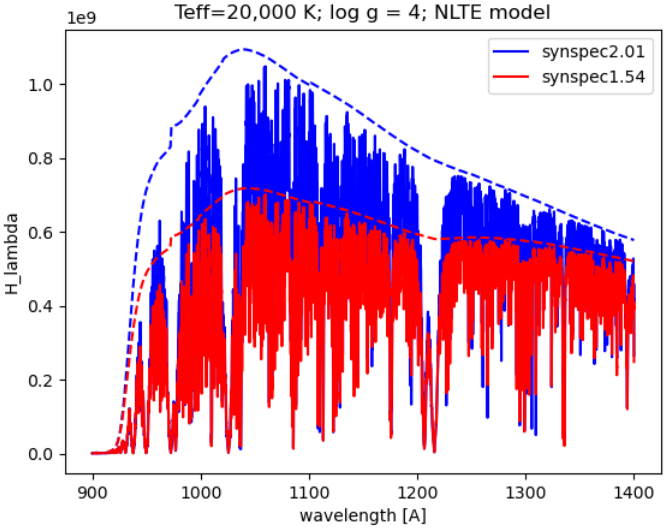


Fig. 4: a synthetic spectrum for a BSTAR07 (Lanz & Hubeny 2007) NLTE metal line-blanketed model atmosphere with $T_{\rm eff} = 20,000$ K; $\log g = 4$, computed with the current version of SYNSPEC, and with SYNSPEC, version 1.54. The plotted spectra are smoothed over 50 points in the original spectra, which corresponds to a spectral resolution of about 0.5 Å. The theoretical continuum is shown as a dashed line.

As mentioned earlier, previous versions of SYNSPEC were not designed to treat molecular ions, such as OH^+, so if lines of those ions were present in the line list, SYNSPEC treated them as lines of a regular neutral molecule, which obviously grossly exaggerated their strength. This is fixed, and

[5] Its creation may however be inhibited by setting a specific keyword parameter – see the Manual. Such possibility is advantageous in the case of computing very extended wavelength ranges of spectrum, since the identification table could become extremely large.

SYNSPEC is now able to deal with essentially any molecular ion.

6.3. Coding issues

As mentioned in the Introduction, SYNSPEC was transformed to FORTRAN90/95. Strictly speaking, not all the available features of the language were used, but the emphasis was placed on producing a more efficient code from the point of view of its use and computer memory requirements. The new elements of the code are described in more detail in the Manual; here we mention the most important features:

1. All COMMON blocks were removed and replaced by MODULES.
2. Most arrays are now allocated at run time, instead of specifying their dimensions in advance. In particular, arrays related to line lists and line atomic data are allocated dynamically, taking into account actual numbers of lines to be dealt with. This allows the user to work with an arbitrary number of individual molecular line lists.
3. The overall appearance of the code has changed; for instance all labels were removed and replaced by specific constructs and format strings.

7. Utility programs

7.1. Construction of isotope-averaged SYNSPEC compatible line lists from EXOMOL data

There are four FORTRAN90/95 programs, which are described in detail in the Manual:

1. PFSUM generates isotope-averaged partition function for a given molecular species, based on the EXOMOL data for the individual isotopologues, using the procedure outlined in § 4.3. The relative abundances of the individual isotopologues are assumed to be solar, with the exact values given in Table 1.
2. LIST extracts line data for a particular isotopologe of a molecule from the EXOMOL data and prints them in the SYNSPEC format. It assumes that the total (isotope-averaged) partition function of the molecule is already given, either computed by a previous run of the program PFSUM, or copied (linked) form some other source. The program works for one isotopologue at a time.
3. REVERSE is a very simple program that takes an output from LIST and reverses the order of lines. The reason is that in the EXOMOL files the line transitions are sorted by increasing wavenumber (decreasing wavelength), while SYNSPEC assumes the opposite ordering, from short to long wavelengths.
4. MERGE is again a very simple program that merges two lists, previously produced by LIST and REVERSE.

When repeating the procedure for all isotopologues of a given molecule, one generates a SYNSPEC-compatible line list for a single molecule.

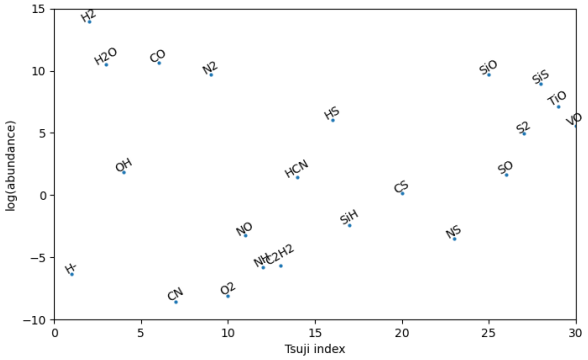


Fig. 5: An example of the solution of EOS using programs CHEMEQ and EOS.PY for $T = 1000$ K, and the total particle number density $N_{\rm tot} = 10^{14}$ cm^{-3}. The x-coordinate is the index of the molecule in the Tsuji data file, which is not important here because the molecules are identified by their chemical labels.

7.2. Equation-of-state program CHEMEQ

The program CHEMEQ is built from several slightly modified or reduced subroutines from SYNSPEC, designed to provide a fast[6] solution of the chemical equilibrium (equation of state) in the presence of all 500 considered molecular species. The program is thus useful for a determination of which molecular species are important for given atmospheric conditions (temperature and density), and therefore helps to decide which molecular line lists are needed for the potential spectrum synthesis using SYNSPEC. The program also provides insight into the question of how the individual atomic nuclei are being sequestered in the atomic and molecular species that contain them as a component, listed in the order of their abundance. Besides guiding the spectrum synthesis, the program may also have a pedagogical value.

The program is accompanied by a short Python code EOS.PY that calls CHEMEQ, and plots the resulting abundances of 92 neutral and ionized atoms and 503 molecular species, for a specified temperature, T, and one state parameter (total particle density). The use of CHEMEQ and EOS.PY is demonstrated on Fig. 5, which shows the abundances of the first 30 molecules in the Tsuji table at $T =$ 1000K, and the total particle number density $N_{\rm tot} = 10^{14}$ cm^{-3}.

7.3. Graphical user interface

Since the input to SYNSPEC is relatively simple and straightforward, we have created two packages that serve as a wrapper to prepare the necessary input files, run the code, possibly perform rotational and instrumental convolutions, and plot the results.

Synplot – an IDL wrapper

SYNPLOT is an IDL wrapper to perform all the tasks outlined above, plus providing line identifications based on the

[6] It takes about 10 ms for one (T/ρ) pair on a MacBook Pro with macOS Sequoia 15.5.

identification table produced by SYNSPEC. The new variant, developed by one of us (Y.O.), works in IDL and GDL[7].

Synple – a Python wrapper

Over the last few years, we have developed a Python code, `Synple`[8]. It is to some extent analogous to SYNPLOT, although there are differences. In the present version, it does not allow to use all functionalities of SYNSPEC, but it is in continuous development.

`Synple` can take as input MARCS, Kurucz, PHOENIX, and TLUSTY model atmospheres, and is designed to be useful to researchers that are not experts in stellar atmospheres and radiative transfer. It facilitates performing massive spectral synthesis calculations for large grids, parallelizing the calculations over multiple cores, using the python libraries `multiprocessing` or `ray`, and multiple processors, setting up scripts to run syn on distributed-memory computers.

It includes as well BAS, the Bayesian Algorithm in Synple, a routine for fitting observed spectra using precomputed grids of synthetic spectra. `Synple` will be described in detail in a forthcoming paper.

8. Summary and outlook

We present a new version of the code SYNSPEC. Compared to the previous versions, it has been transformed to FORTRAN90/95, which among other advantages allows for a more efficient handling of atomic and molecular data. In particular, besides providing an updated treatment of molecular partition functions and equilibrium constants, it enables working with an arbitrary number of separate molecular line lists, and having an individual line list for every molecular species. Because of this fundamental coding change, we abandoned the previously used numbering of versions (ending with last released version 54), and adopted a new numbering, starting with the present version 2.01.

We also present a suite of utility programs. One of these has been designed to generate SYNSPEC compatible line lists directly from the EXOMOL database. The users will then. be able to generate various tailored line lists for specific applications. Another utility program is CHEMEQ, and its companion Python wrapper EOS.PY, designed to provide a fast solution of the equation of state, and graphically displaying its results. Two more utilities, SYNPLOT and `Synple`, provide a user-friendly command-line interface and interactive graphics in IDL/GDL and Python, respectively.

The present paper is devoted to describing the underlying physics adopted in SYNSPEC, without discussing practical issues such as a description of input and output files. This is done in an accompanied document, referred here as the Manual. This document, together with the program SYNSPEC, the utility programs, a directory with all necessary atomic and molecular data, and a description of test cases, is distributed through GitHub, at

`https://github.com/ivanek-hub/synspec2.01`

In the future release of SYNSPEC, we plan to incorporate several physical improvements. We will incorporate all available, present and future, EXOMOL data for partition functions and line transition data, including those relevant for very cool temperatures, such as CH_4 and NH_3, which both have a vast number of lines. We will also combine the EXOMOL partition functions and BC dissociation energies to construct new equilibrium constants for molecules of interest.

We will incorporate into a forthcoming release new theoretical, quantum-mechanical data for hydrogen line profiles, including contributions from transient H^- and H_2^+ molecules, sometimes called quasi-molecular satellites. Finally, we will make fully functional a collection of existing routines for spectrum synthesis in spherically-symmetric stellar atmospheres with macroscopic velocity fields.

The upgrades described in this paper make of the new SYNSPEC a powerful tool for research across many fields of astrophysics. The code and its wrappers enable massive calculations of spectral libraries with a modest effort, and such libraries will fuel the analysis of data from ongoing and upcoming massive spectroscopic surveys.

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[7] `https://owncloud.iac.es/index.php/s/0CO8xQYTZXjJpxN`
[8] https://github.com/callendeprieto/synple